ARCHITECTURE REVIEW PRESENTATION

# Hostel Buddy

## Smart Hostel Complaint Management System

**Group 19**

Pranjal Tyagi · Ajinkya · Dinesh Saini · Saurabh · Vijay

Department of Computer Science and Engineering · **IIIT Kottayam**

# Agenda & Speaker Allocation

20 minutes

| # | Topic | Time | Presenter |
|---|---|---|---|
| 1 | Project Overview | 1 min | **Pranjal Tyagi** |
| 2 | Module Decomposition | 2 min | |
| 3 | Activity Diagram | 1.5 min | |
| 4 | System Architecture | 2.5 min | **Ajinkya** |
| 5 | Database Design — ER Diagram | 1.5 min | |
| 6 | Class Diagram | 1.5 min | **Dinesh Saini** |
| 7 | Layered Architecture & Data Flow | 2.5 min | |
| 8 | Sequence Diagram — with failure paths | 1.5 min | **Saurabh** |
| 9 | Security Controls, Design Decisions & **Accepted Trade-offs** | 2 min | |
| 10 | State Diagram | 1.5 min | **Vijay** |
| 11 | System Requirements & Use Case Diagram | 2 min | |
| — | Conclusion | 0.5 min | **All** |

# 1 Project Overview

Pranjal Tyagi · 1 min

### Problem

- Hostel complaints are recorded in a **paper register**
- No way to search, track or report on them
- Students get **no visibility** after reporting an issue

### Objective

- Digitise complaint reporting and resolution
- Give students **real-time status tracking**
- Give the administrator a **single dashboard** with analytics
- Store every complaint **permanently and searchably**

### Who uses the system

- **Student** — raises and tracks their own complaints
- **Administrator** — reviews all complaints, updates status, views analytics

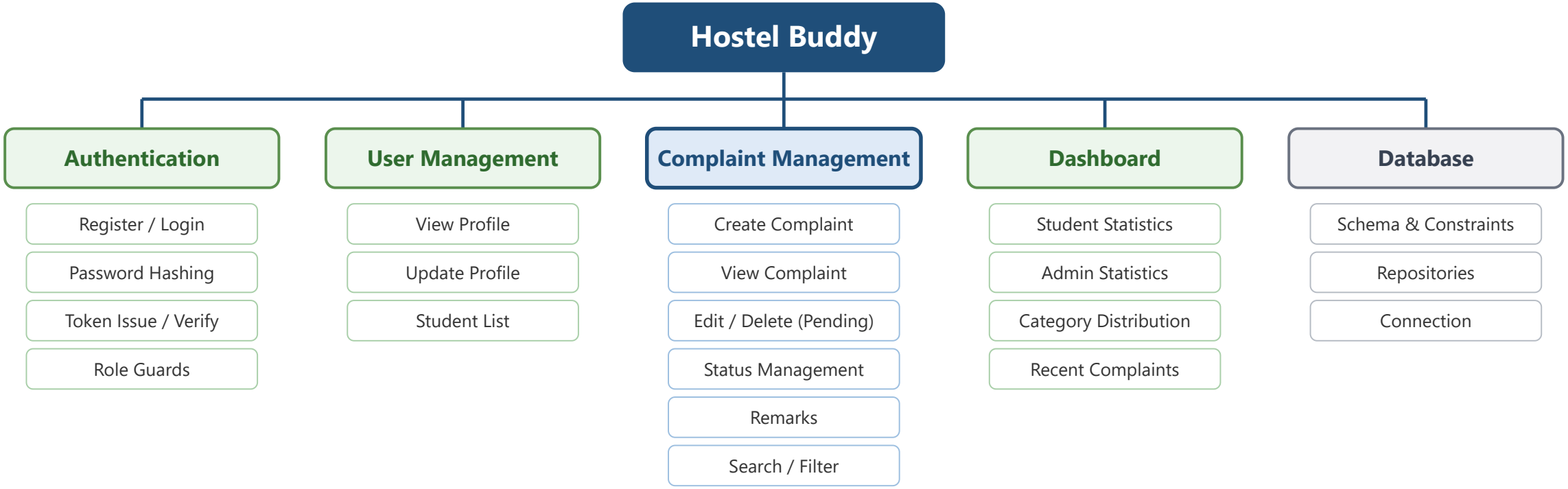
Two roles only — no separate maintenance-staff role in v1.0

### How it works — in one line

A student submits a complaint through a web interface; it is stored as **Pending**, and the administrator advances it through **In Progress** → **Resolved** → **Closed** while the student watches the status change live.

# 2 Module Decomposition

Pranjal Tyagi · 2 min

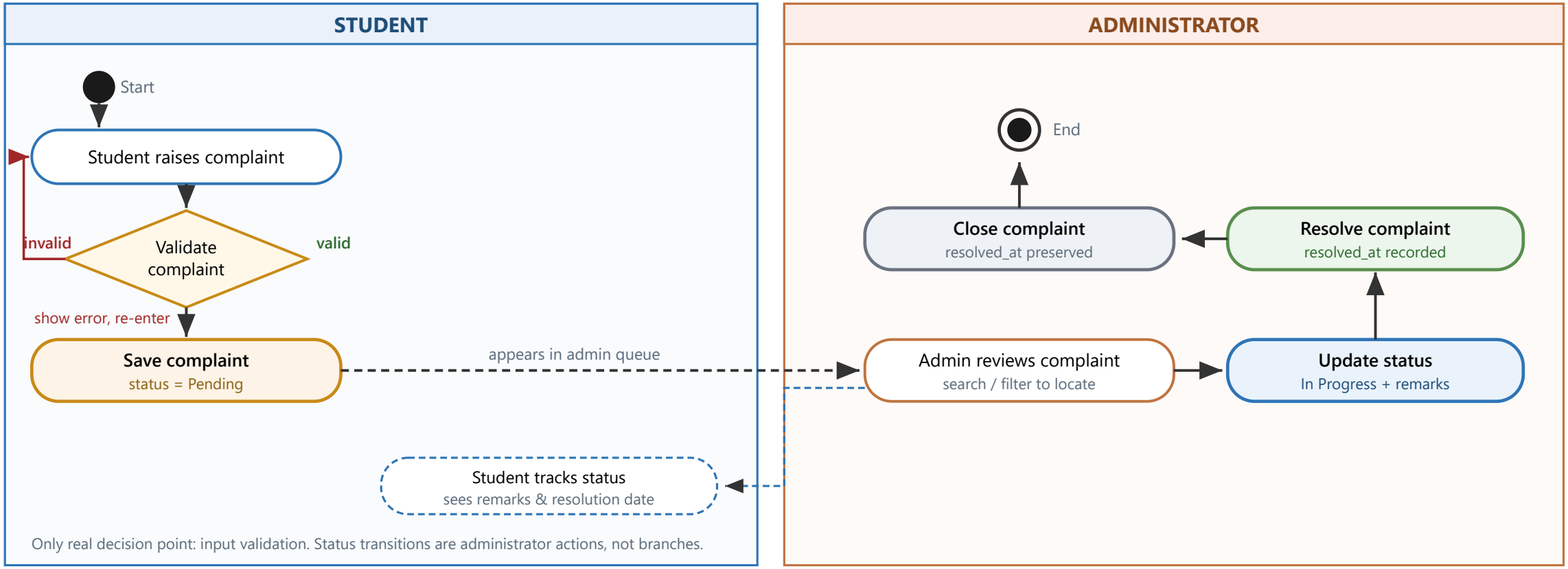



**Purpose of decomposition**

Each module has one specific responsibility — this makes the system easier to **maintain**, **test** and **extend**.

# 3 Activity Diagram — Complaint Process

Pranjal Tyagi · 1.5 min



# 4 System Architecture — Three-Tier

Ajinkya · 2.5 min

## Three-Tier View

**Presentation Tier**
Student / Admin → Frontend
HTML · CSS · JavaScript · Chart.js

↓ JSON over HTTPS + JWT

**Application Tier**
Express API
requests · business logic · authorisation

↓ SQL (prepared statements)

**Data Tier**
SQLite
users · complaints + /uploads

**Responsibilities**

- Presentation — handles the UI
- Application — handles requests & business logic
- Data — stores persistent data

## Inside the Application Tier

**Routes**
map URL + verb to a handler

↓

**Middleware**
authentication · role check · upload

↓

**Controllers**
parse request · shape response

↓

**Services**
ALL business rules live here

↓

**Repositories**
the only layer that runs SQL

↓

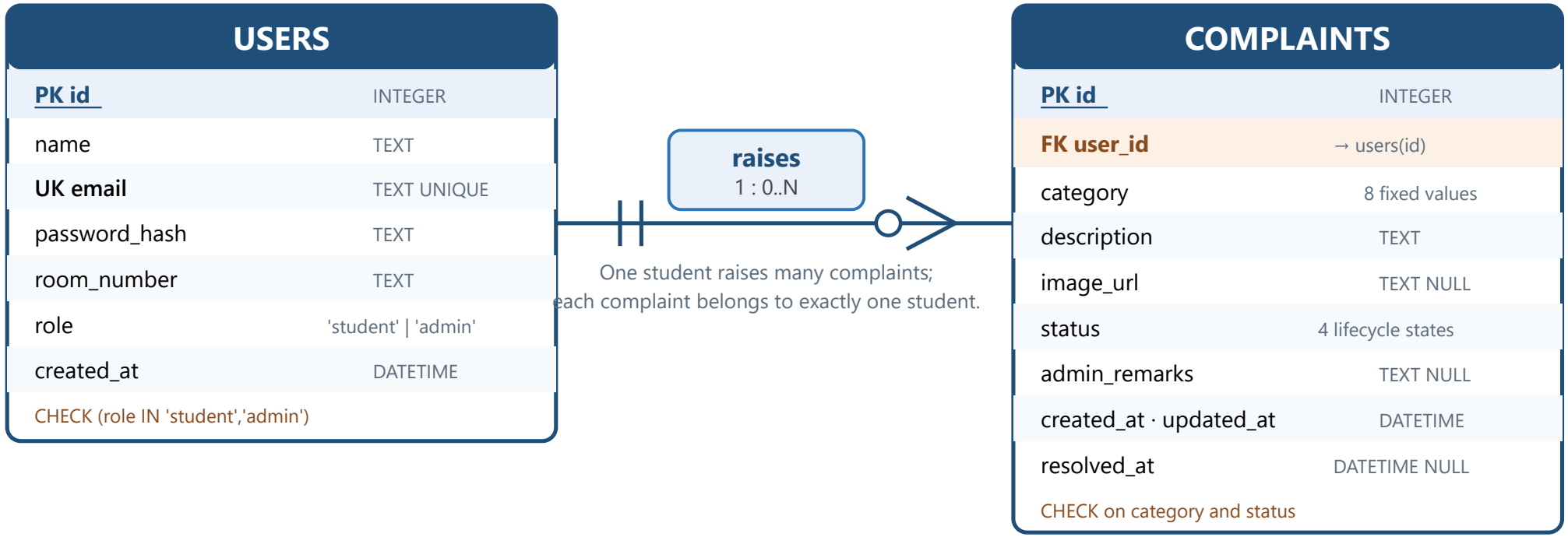
**SQLite**
persistent storage

Each layer talks only to the layer directly below it.

Business rules never appear in the frontend or in SQL.

# 5 Database Design — ER Diagram

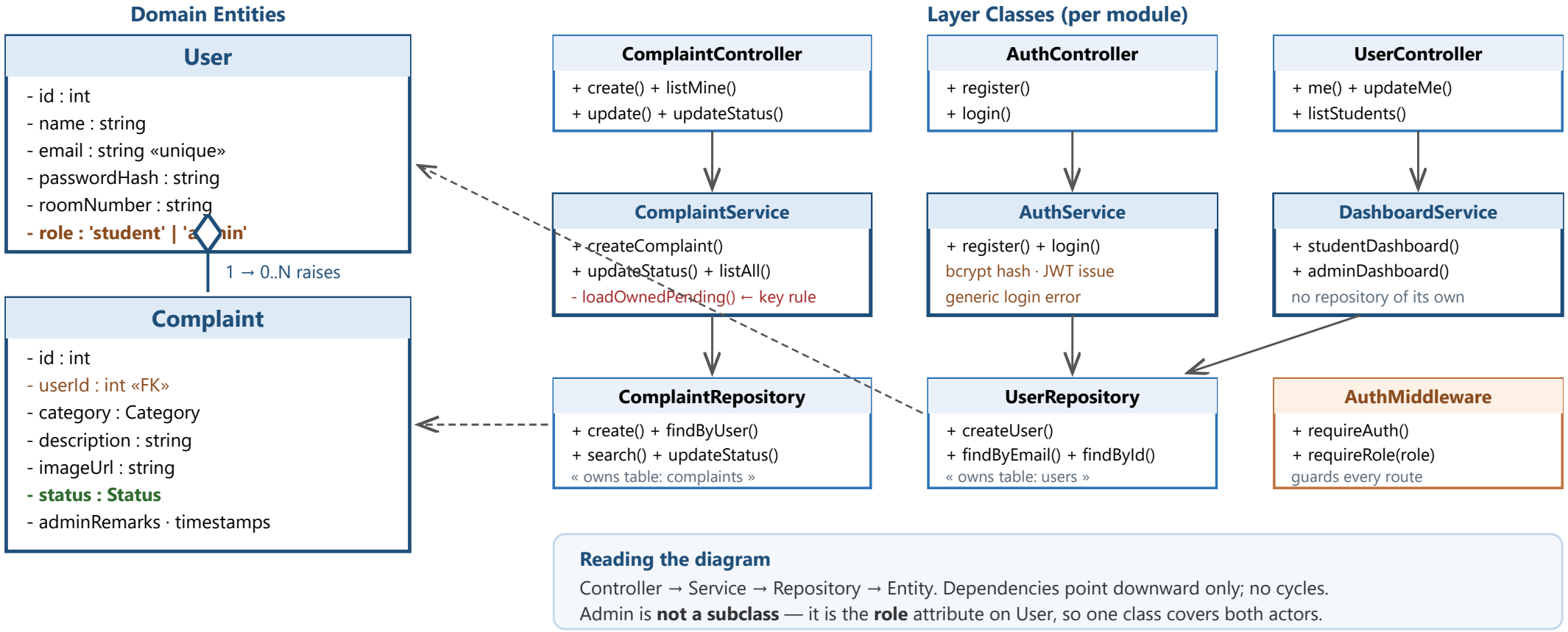
Ajinkya · 1.5 min




**Two entities only.** Admin is a **role** on USERS, not a separate table — authentication stays uniform.
Categories and statuses are enforced by **CHECK constraints** rather than lookup tables — fixed sets, one fewer join.

# 6 Class Diagram

Dinesh Saini · 1.5 min




**Reading the diagram**

Controller → Service → Repository → Entity. Dependencies point downward only; no cycles.
Admin is **not a subclass** — it is the **role** attribute on User, so one class covers both actors.

# 7 Layered Architecture & Data Flow

Dinesh Saini · 2.5 min

## Layer Responsibilities

**Routes**
Decide which handler and which guards apply.

↓

**Middleware**
Authenticate the caller, check role, handle upload.

↓

**Controllers**
Read the request, call a service, return a response.

↓

**Services**
Apply the business rules — validation, ownership, lifecycle.

↓

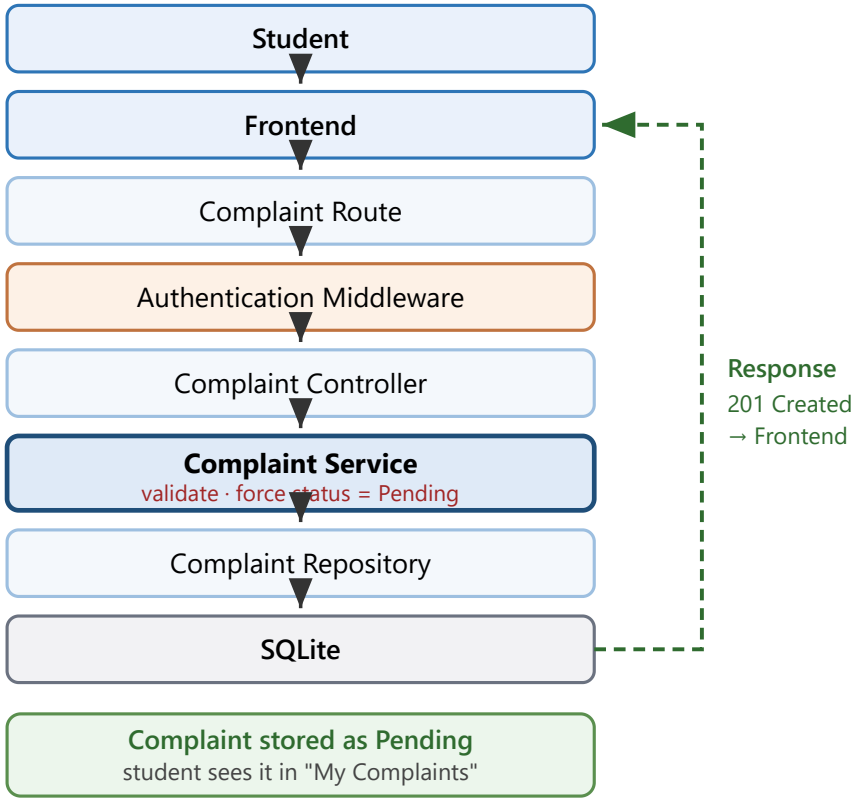
**Repositories**
Execute SQL and return plain objects.

↓

**Database**
Store the data durably with constraints.

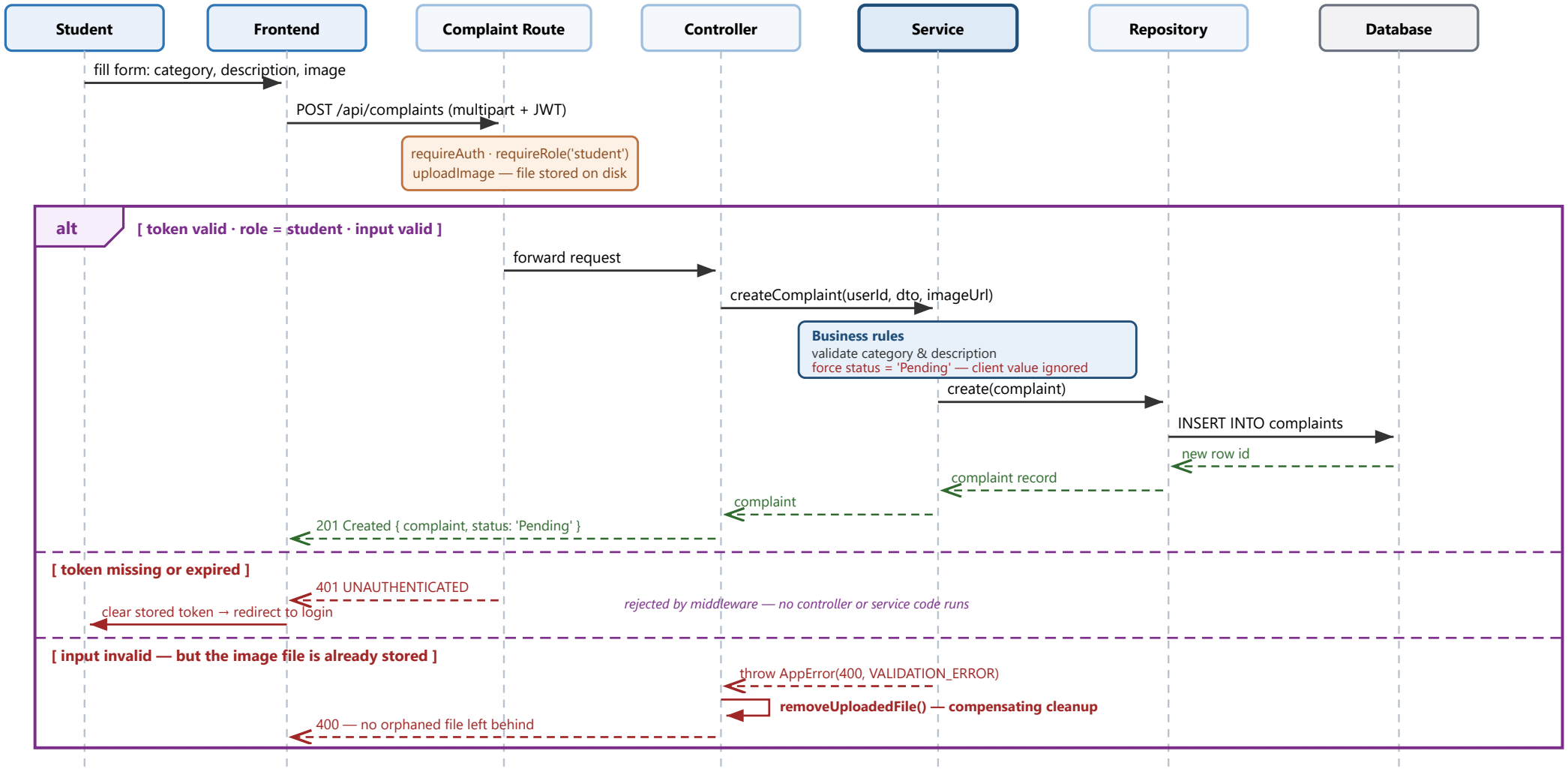
A request can never skip a layer.

## Concrete Request — Raise Complaint



# 8 Sequence Diagram — Raise Complaint, with Failure Paths

Saurabh · 1.5 min




**Fourth branch — `PUT /complaints/:id` where status ≠ Pending:** the **Service** (not the Route) throws `409 NOT_PENDING` — a route-level check would protect only that one route, a service invariant holds for **every** caller.

# 9 Security Controls & Key Design Decisions

Saurabh | 2 min (9)

## Security Controls

- **JWT authentication**
  stateless signed tokens · 24-hour expiry
- **bcrypt password hashing**
  salted, cost factor 10 · never reversible
- **Role-based access control**
  route guard + service ownership check
- **Helmet / CSP**
  security headers · scripts restricted to our origin
- **Rate limiting**
  on authentication endpoints, in production

## Why SQLite?

- Built into Node.js
- No compiler / native build requirement
- Simple setup for the model version

## Why layered architecture?

- Separation of concerns
- Easier maintenance and testing
- Business logic independent from database access

## Why a Repository layer?

- Keeps SQL / database operations isolated
- Makes future database replacement easier

# 9 Accepted Trade-offs

Saurabh · 2 min (9)

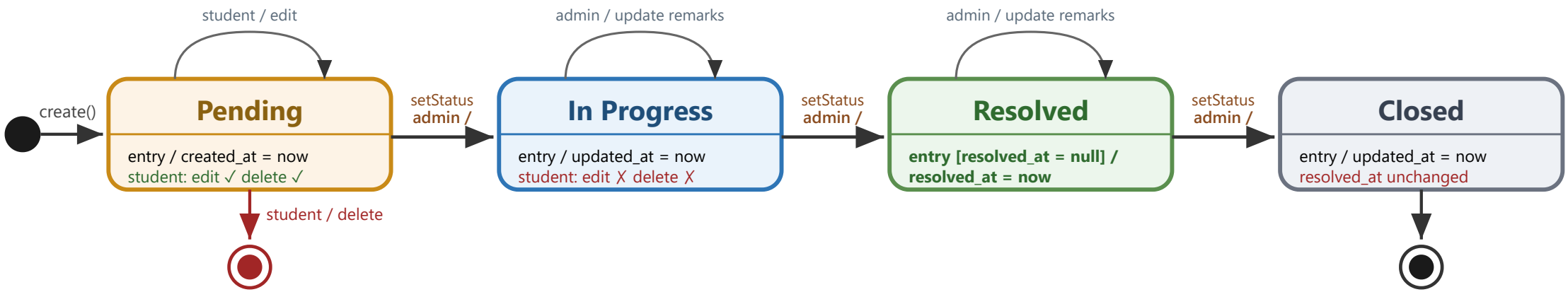
Three limitations we chose **knowingly** — each with a bounded risk and a documented upgrade path.

| Weakness we accepted | Why we accepted it | What bounds the risk |
|---|---|---|
| **JWT cannot be revoked before it expires** | Statelessness is what lets us replicate instances horizontally — a session store would reintroduce shared state | 24-hour token lifetime caps the exposure window; a server-side deny-list or refresh-token scheme attaches at the **existing verification middleware** |
| **Complaint images are served unauthenticated** | Gating every image means streaming each one through an authorised route — real complexity for little gain at this scale | Filenames are **timestamp + random hex**, so they are impractical to guess, and URLs appear only inside access-controlled views |
| **SQLite allows only a single writer** | Complaint reads vastly outnumber writes; a separate database server adds a network hop and another failure mode | WAL mode preserves read concurrency; the PostgreSQL migration touches **only the repository layer**, never business logic |

Every trade-off is isolated behind a single layer — which is what makes each one reversible.

# 10 State Diagram — Complaint Lifecycle

Vijay · 1.5 min




### Lifecycle rules enforced by the system

- The student **creates** the complaint — it always starts in **Pending**; the **admin controls every transition**.
- **The student cannot change status** — there is no student operation that writes it.
- **resolved_at is recorded at Resolved**, once only — moving to Closed never overwrites it.
- Edit / delete exist **only in Pending**; a state outside these four is rejected by a database CHECK constraint.

# 11 System Requirements

Vijay | 2 min (11)

## Student Functional Requirements

- Register / Login
- Manage profile
- Raise complaint
- View complaints
- Edit / Delete **Pending** complaint
- Track complaint status
- View remarks and resolution date
- View personal statistics

## Administrator Functional Requirements

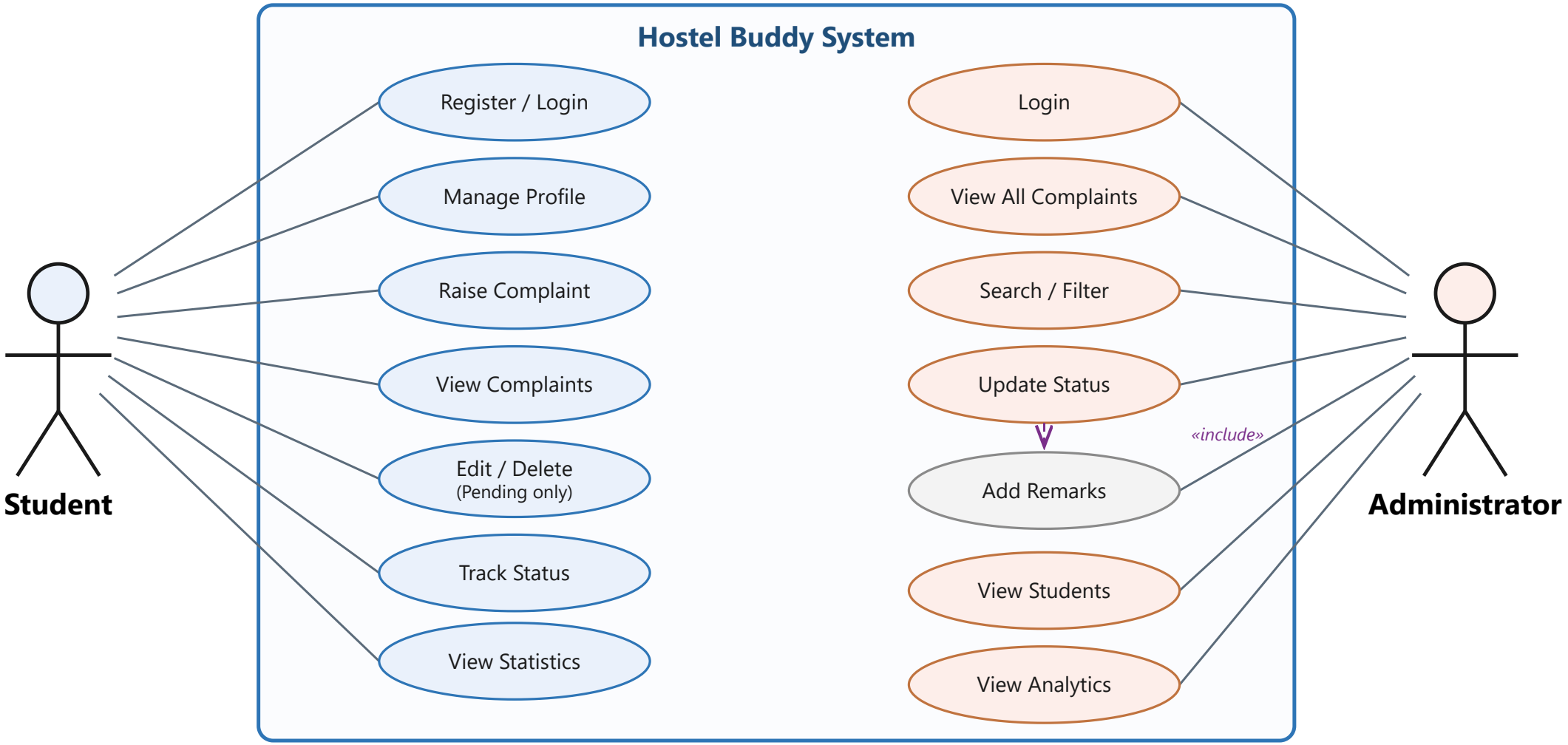
- Login
- View all complaints
- Search / filter complaints
- Update complaint status
- Add remarks
- View registered students
- View analytics

## Non-Functional Requirements

- **Security** — hashed passwords, role-based access
- **Usability** — responsive, works on mobile
- **Maintainability** — independent modules
- **Reliability** — no complaint data loss

# 11 Use Case Diagram

Vijay | 2 min (11)




Two actors only. Administrator drives every status transition; the student is read-only on status.

# ✓ Conclusion

0.5 min

1. Hostel Buddy uses a **modular layered architecture** — four feature modules over a shared infrastructure layer.
2. **Business logic is separated** from presentation and database access — services hold the rules, repositories hold the SQL.
3. The **complaint lifecycle is controlled by authorization rules** — only the administrator advances status; students edit only while Pending.
4. The architecture is **maintainable and suitable for the current model version**, with a clear path to scale.

Thank you — questions?